

VSAE ECONOMETRIC GAME 2023

Predict the Hotspots of Child Malnutrition

19, 20 and 21 April 2023

Abstract

Background. With population growth that has exploded since the 1990s, Africa will probably become the most populous continent in the world in 2050. But it is also the continent most affected by malnutrition. Obviously a social issue, but also an economic and political one, this is the subject that this paper deals with. Notably for geographical reasons, Sahel is the region most affected by this scourge. The main purpose of this study is to analyse child malnutrition, and predict the indicator called burden of wasting.

Methods. In the first place, we applied a Lasso and an Elastic-net to do dimensionality reduction. In order to predict the level of child malnutrition, we implemented several supervised machine learning models for classification problems: a Random Forest, a Support Vector Machine, a K-Nearest Neighbors and an Extreme Gradient Boosting classifiers.

Results. We compared the performance of these models using the Mean Squared Error, the Accuracy and the ROC curve (with precision and recall). Our study yields that the XGBoost classifier produces the best prediction, with an accuracy of 89.5%. The second best is the Random Forest (86.6%), then the K-NN (86.1%) and finally the SVM classifier (83.6%).

Conclusions. The best algorithm to predict the level of child malnutrition in regions of sub-Saharan Africa is XGBoost. The features with the highest weight in the prediction are the access to health care, food insecurity, internally displaced persons, and death rate associated with malaria.

Contents

1	Introduction	3
2	Materials and Methods	5
2.1	Data presentation and preprocessing	5
2.2	Description of the methodology	6
2.2.1	Dimensionality reduction by Lasso	6
2.2.2	Dimensionality reduction by Elastic-net	7
2.2.3	Random Forest classifier	7
2.2.4	Support Vector Machine	8
2.2.5	K-Nearest Neighbors classifier	8
2.2.6	eXtreme Gradient Boosting	9
2.2.7	Shap values	9
3	Results	10
3.1	Exploratory analysis	10
3.1.1	Malnutrition levels	10
3.1.2	Explanatory variables	11
3.1.3	Exploring Correlations Between Explanatory Variables and Malnutrition Levels	12
3.2	Predictive analysis results	13
3.2.1	Dimensionality reduction	13
3.2.2	Machine learning algorithms	14
3.2.3	Model selection	16
3.2.4	SHAP Values for XGBoost	17
4	Conclusion	21
5	References	22

1 Introduction

Malnutrition is a significant public health problem that has disastrous consequences for child survival and healthy development. According to information provided by UNICEF, a child can have several episodes of acute malnutrition in the same year, and according to the joint UNICEF, WHO, and World Bank estimates on malnutrition (2020 edition), 7.3 million children were suffering from wasting (severe and moderate) in West and Central Africa in 2019. However, for decades, some countries in the region have managed to deal with the scourge of child malnutrition while others have struggled to find the right formula.

According to specific surveys conducted by the Population Reference Bureau, nutrition is generally influenced by five interconnected factors: political instability, poverty and inequality, ineffective development policies, environmental and climate change, and inadequate and poorly managed food security, nutrition, and health prevention programs. Moreover, these factors remain indeed non-exhaustive. This problem of child malnutrition increases the mortality rate every year in the world. When we focus on specific regions worldwide, we see that much remains to be done. If this problem cannot be correctly tackled, the authorities need to properly define the factors that determine the level of malnutrition in a country. The most crucial concern remains to discover the factors determining malnutrition in a given country or region.

To address this issue, some authors have conducted studies on malnutrition employing various techniques. However, only a few studies have focused on predicting the level of malnutrition and the key factors that can predict the level of malnutrition in a given country. The following literature review covers studies exploring different machine-learning algorithms for predicting malnutrition in different countries and regions.

For instance, research by Islam et al. (2022) discovered that machine learning algorithms such as Naïve Bayes, support vector machine, decision tree, artificial neural network, and random forest were able to develop prediction models for malnutrition among women in Bangladesh. Their findings revealed that the Random Forest-based classifier had the highest accuracy (81.4%).

Bitew et al. (2021) employed data from the Ethiopian Demographic and Health Survey of 2016 to exhibit the relative superiority of machine learning models in predicting undernutrition among under-five children in Ethiopia. They considered eXtreme gradient boosting, k-nearest neighbors (K-NN), random forest, neural network, and generalized linear models, where the xgbTree algorithm demonstrated superior performance with an accuracy of 88%. Additionally, they highlighted the need for improvement in access to water supply, food security, and fertility regulation, as they were the most relevant features in the prediction.

Kar et al. (2021) focused on malnourished children aged 1 to 5 in Asia, utilizing four ML classifiers, with the best results achieved by Random Forest and Logistic Regression, yielding accuracies of 91.11% and 89.88%, respectively.

Khan et al. (2023) examined stunting, underweight, and wasting of children under five in sub-Saharan African countries using random forest, decision tree, eXtreme Gradient Boosting, and k-nearest neighbors ML methods. The study reported accuracy values of 81% for the random forest, 60% for the decision tree, 79% for eXtreme Gradient Boosting, and 74% for k-nearest neighbors. They found the most significant features to be Mother’s age, the child’s fever, and education level.

Fenta et al. (2021) explored the undernutrition status of under-five children in Ethiopia using six ML algorithms: Logistic regression, Least Absolute Shrinkage and Selection Operator, Elastic net, neural network, and random forest. The RF algorithm was selected as the best ML model based on performance evaluations (67% of accuracy). The study identified urban-rural settlement, literacy rate of parents, and place of residence as significant determinants of disparities in nutritional status.

Talukder et al. (2020) investigated malnutrition status among children in Bangladesh using five ML algorithms: linear discriminant analysis, k-nearest neighbors, support vector machines, random forest, and logistic regression, with the RF algorithm demonstrating the best results in accuracy at 68.51%.

These studies have shown that machine learning algorithms can accurately predict malnutrition status among different populations using various features. Random forest, eXtreme Gradient Boosting, and k-nearest neighbors are commonly used algorithms that have achieved high accuracy results in specific contexts. Accuracy rates across the studies varied; some achieved rates above 80% while others ranged from 60-70%. Relevant features used to predict malnutrition included access to clean water, food security, literacy rate of parents, mother’s age, and child’s fever, among others.

However, while there is a growing body of literature on the use of machine learning techniques to improve the timing of interventions of public policies, these studies focus on countries that might have different characteristics than the G5 Sahel countries, such as poorer living and socio-economic conditions. This represents a gap in the literature, as the aspects of the regions can significantly affect the accuracy of the targeting. Therefore, our study aims to fill this gap by analyzing the case of the G5 Sahel countries. There is also a need for interpretability in the literature, as it is crucial for policymakers to understand the most relevant factors in determining early identification. Our study addresses this need by using SHAP values, which provide interpretability and assess the impact of each variable on the prediction to ensure a better understanding of the model and its results.

To address the question of forecasting the level of malnutrition for children aged between 6-59 months in the G5 Sahel (plus Senegal), we employ an empirical analysis using machine learning techniques. To conduct our research, we use a dataset from the nutritional surveys from the G5 Sahel region plus Senegal, and from the publicly available INFORM dataset from 2021¹. Our approach includes data collection and preprocessing, dimensionality reduction using Lasso and Elastic-net, application of machine learning models such as Support Vector Classifier, Random Forest Classifier, K-Nearest Neighbors, and eXtreme Gradient Boosting, and model evaluation using prediction accuracy. Our goal is to compare these models to assess which one has the best performance, which is crucial for policymakers in determining the most efficient allocation of resources. We also use SHAP values to provide interpretability and assess the impact of each variable on the prediction. By employing this empirical analysis, we can develop models that can accurately predict the level of malnutrition for children aged between 6-59 months in the G5 Sahel region, which is critical for efficiently allocating resources to those most in need. This approach allows us to provide a comprehensive and robust analysis of the problem and draw meaningful conclusions and recommendations for policymakers.

In this study, we found several key results that provide valuable insights for policymakers and practitioners in the field of malnutrition identification that could lead to early interventions. Our results showed that using machine learning techniques significantly improved the prediction of the level of malnutrition in children aged between 6-59 months. The eXtreme Gradient Boosting model performed best in terms of prediction accuracy. Additionally, SHAP values provided interpretability and showed that variables related to access to health care, food insecurity, internally displaced persons, and the death rate associated with malaria were among the most critical factors for the level of malnutrition prediction. Our findings have the potential to provide valuable insights into this field, as well as for other public policies that aim to target individuals or households based on their characteristics. Ultimately, our goal is to contribute to the ongoing effort to allocate resources efficiently and mitigate child malnutrition.

This study proceeds as follows. Section 2 introduces an overview of the Dataset used in the analysis and describes the techniques and algorithms employed to analyze the data. Section 3 presents the results, an exploration of the data, and the results of the predictive models. Finally, section 4 summarizes the study’s main findings and discusses their implications.

¹Dataset available here: INFORM

2 Materials and Methods

2.1 Data presentation and preprocessing

The data used in this work come mostly from nutritional surveys from the Sahel region and a publicly available INFORM dataset². The goal is to predict the burden (Burden = Prevalent Cases+Incident Cases) of wasting or the level of malnutrition. This dataset provides 69 variables with the dependent variable and 1203 observations. Our target variable has four modalities: Low, Medium, High, or Very High levels of malnutrition.

Based on these variables and observations, it will therefore be a question of finding the variables that best predict the level of malnutrition according to the country and its socioeconomic/socio-demographic situation. To achieve our goal, we proceeded to clean the database because the initial database contains many missing values, which, if not well treated, can bias the estimation results. To do this, we first calculated the percentage of missing values per variable and deleted the variables with a percentage of missing values strictly greater than 50%. This approach allowed us to reduce the database by seven variables, which can be explained mainly by other variables in the database. So we do not lose enough information by deleting them. Secondly, we note that there are countries such as Nigeria for which we have no information on the three years, and several variables have no value. It is also the case for Sierra Leone, where we have no values for several variables. To not assign the values of other countries to another, we decided to remove these two countries so that our results reflect reality better.

After this cleaning, we have a database of 62 variables and 1125 observations. This deletion of the two countries confirms the choice made in the statement, which asks us to consider the G5 countries plus one more country after cleaning: Senegal.

To deal with the few other missing values in the database, we have chosen the hot-deck method among all the existing methods based on the K nearest neighbors. In Kwon et al. (2014), the authors test several hot deck imputation methods to choose the best one. We found the hot deck method using K-NN, deletion of listwise, mean, mode, linear regression, and SVM imputation among them. After analysis, they found that the K-NN hot deck method gave the best performance. Indeed, according to Joensuu and Bankhofer (2012), hot deck methods impute missing values within a data matrix by using available values from the same matrix. The donor is the object from which these available values are taken for imputation within another. The replication of values leads to the problem that a single donor might be selected to accommodate multiple recipients.

²Dataset available here [INFORM](#)

2.2 Description of the methodology

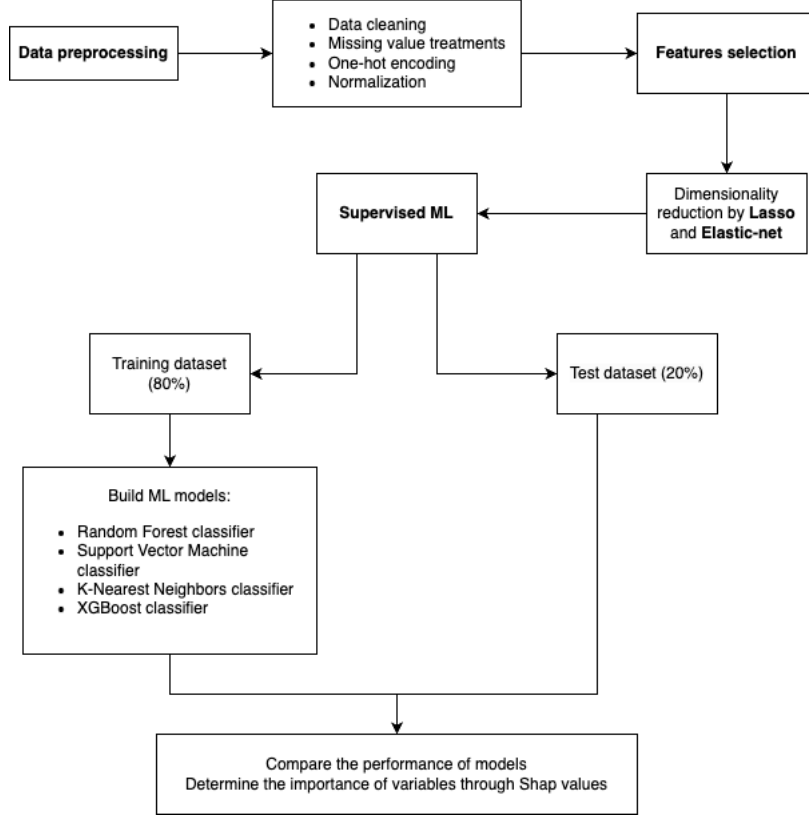


Figure 1: Methodology

Figure 1 illustrates the methodology employed to predict child malnutrition. The process started with data preprocessing, which involved data cleaning, missing value imputation using Hot-Deck with KNN, one-hot encoding, and normalization. Subsequently, dimensionality reduction was carried out using Lasso and Elastic-Net models to select relevant features. The preprocessed dataset was then partitioned into training and testing sets. Four machine learning models, namely Random Forest classifier, Support Vector Machine classifier, K-Nearest Neighbors classifier, and XGBoost classifier, were constructed and their performance was evaluated. These models were selected accordingly to what is being done in the most recent literature. Shap values were employed to enhance the interpretability of the models. Further details on the methods used will be elaborated on in the subsequent section.

2.2.1 Dimensionality reduction by Lasso

It is crucial to note that the database used in this study comprises a substantial number of potential features. A high number of variables can entail a significant computational burden and may introduce noise by including variables in the model that do not significantly contribute. To address this, shrinkage methods are employed, given that they allow for fitting models with all predictors while incorporating constraints or regularizations that shrink certain estimated coefficients toward zero. James et al. (2021) state that the most well-known methods in this regard are Ridge and Lasso. These methods aim to reduce the variance and enhance the model's fit.

In contrast to Ridge, Lasso's method seeks to force some estimated coefficients to be precisely zero when the penalty parameter is sufficiently adequate. In contrast, Ridge's method sinks certain coefficients toward zero without reaching an exact zero value. Consequently, Lasso performs a variable selection in which it obtains coefficients β_{λ}^L that can be obtained from the following minimization (James et al., 2021):

$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j| = RSS + \lambda \sum_{j=1}^p |\beta_j|$$

The minimization involves the selection of a penalty parameter " λ ". The choice of " λ " is of paramount importance, as a very low parameter will penalize only a minimal number of coefficients. At the same time, a very high value will yield a null model in which all coefficients are zero. According to James et al. (2021), this parameter choice must be made by the smallest cross-validation error. First, by choosing a grid of possible values for the parameter, and then computing the cross-validation error of each value.

During the execution of this selection of variables, it was not only important to use the models, but also GridSearchCV was used. GridSearchCV seeks to select the best and most optimal parameters based on a grid that considers different possible values for the Lasso hyperparameters. The selection of the parameters works with cross-validation, a method that seeks to divide the data into subgroups (k-folds), k-1 folds are used to train the model, and 1 is used to test it. In cross-validation the process is repeated several times testing all the combinations of test/train sets and the average error is used as the indicator of optimization since it is desired that this is the smallest possible. The hyperparameters that from cross-validation obtain the smallest errors will be selected, which shows that they are the best values to model the data.

2.2.2 Dimensionality reduction by Elastic-net

The Elastic Net dimensionality reduction method is a combination of both the Lasso and Ridge methods. Specifically, the Elastic Net modifies the minimization, incorporating a penalty term in the following manner:

$$RSS + \lambda \left(\sum_{j=1}^p (\alpha \beta_j^2 + (1 - \alpha) |\beta_j|) \right)$$

Hastie (2005) stated that the Elastic Net aims to select variables like Lasso and shrinks the coefficients of correlated variables like the Ridge method. Similar to the Lasso method, the Elastic Net also involves selecting the penalty parameter " λ ", but additionally requires the selection of the hyperparameter " α ". Specifically, " α " defines the degree to which the penalty term is a mixture between Ridge and Lasso. In simpler terms, with the Elastic Net method, a reduction similar to Lasso can be achieved by setting " α " equal to 0, and a reduction similar to Ridge can be achieved by setting " α " equal to 1.

For elastic-net, GridSearchCV is also used as a hyperparameter optimization method. The difference with respect to Lasso is that while Lasso is only looking for one hyperparameter, the elastic-net method seeks to optimize two hyperparameters.

2.2.3 Random Forest classifier

To understand the intricacies of a Random Forest classifier, it is necessary to first familiarize oneself with tree-based methods, which are useful for both classification and regression problems. In this analysis, we focus on using classification trees with the goal of predicting the target class. The construction of regions for a classification tree is achieved through recursive binary splitting with the aim of minimizing impurity measures such as the Gini index or entropy. For each observation in a given region, we predict that it belongs to the most commonly occurring class of training observations in that region. The classification error rate is simply the fraction of training observations in a region that do not belong to the most common class (James et al., 2021):

$$E = 1 - \max_k (\hat{p}_{mk})$$

Here, $\hat{p}_{mk}$ represents the proportion of training observations in the m th region that are from the k th class. The output of the Random Forest for classification tasks will be the class selected by most trees. To execute this model, we also utilize GridSearchCV, as explained above.

2.2.4 Support Vector Machine

The Support Vector Machine (SVM) is an extension of the Support Vector Classifier that seeks to expand the feature space to accommodate non-linear boundaries between classes, thereby enabling the identification of the class to which each observation belongs (James et al., 2021). To comprehend the SVM, it is essential to begin with the Maximal Margin Classifier, considered a classifier that separates observations using a hyperplane (a flat affine subspace). The selection of the optimal hyperplane is based on identifying the observations close to the boundary and creating support vectors, then considering the distance to those vectors as the margin. In this vein, the Support Vector Classifier, also referred to as the soft margin classifier by James et al. (2021), seeks to find the smallest possible margin while allowing some observations to be on the wrong side of the margin and even on the incorrect side of the hyperplane.

The SVM utilizes non-linear boundaries by enlarging the feature space through kernels. The concept of using kernels is to quantify the similarities between observations; thus, kernel selection allows for flexibility in the decision boundary. As an example, using kernels for a two-dimensional dataset results in a non-linear function such as:

$$f(x) = \beta_0 + \sum_{i \in S} K(x, x_i)$$

The execution of the SVM model also requires the use of GridSearchCV.

2.2.5 K-Nearest Neighbors classifier

The K-Nearest Neighbors (KNN) classifier is linked to the concept of instance-based learning. In contrast to model-based learning methods that attempt to learn a general function from the data, instance-based learning stores the training data and uses it to make predictions for new instances based on their similarity to the stored data (James et al., 2021).

The KNN classifier is a non-parametric instance-based learning method, which predicts the class of a new observation by considering the majority class among its k nearest neighbours in the feature space. The distance metric, such as Euclidean distance, is commonly used to measure the similarity between instances.

Formally, given a new observation x_0 , the KNN classifier assigns it to the class with the highest frequency among its k nearest neighbors:

$$\hat{Y}(x_0) = \frac{1}{k} \sum_{i \in N_0} I(y_i = j)$$

where N_0 is the set of the k nearest neighbors of x_0 , y_i is the class label of the i -th nearest neighbor, and $I(y_i = j)$ is an indicator function that equals 1 if $y_i = j$ and 0 otherwise.

The selection of the appropriate value for k has a significant impact on the performance of the classifier. A small value of k may result in a highly flexible model that is sensitive to noise, while a large value of k may lead to a smoother decision boundary that could potentially underfit the data. The optimal value of k can be found using techniques such as K-folds cross-validation, with the objective of selecting the value that minimizes the mean error.

In addition to selecting the optimal value of k , other hyperparameters, such as the distance metric and weighting scheme, can be tuned to improve the performance of the classifier. To search for the optimal hyperparameters, GridSearchCV can be employed, as previously explained.

2.2.6 eXtreme Gradient Boosting

The eXtreme Gradient Boosting (XGBoost) classifier is an advanced implementation of gradient boosting machines, which is an ensemble method that combines several weak learners to create a strong learner. Gradient boosting is an extension of the boosting methods, and in the case of XGBoost, the weak learners are decision trees. The underlying idea of gradient boosting is to iteratively build weak learners by fitting them to the residuals of the previous learners, effectively reducing the errors in each iteration (James et al., 2021).

The XGBoost classifier has several hyperparameters that influence its performance, such as learning rate, maximum tree depth, number of estimators, and regularization parameters. Tuning these hyperparameters, as James et al. (2021) emphasize for boosting methods, can greatly affect the performance of the model.

The objective function of the XGBoost classifier can be expressed as follows:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k k\Omega(f_k)$$

where $l(y_i, \hat{y}_i)$ is the loss function measuring the difference between the true labels y_i and the predicted labels $\hat{y}_i$, f_k represents the k -th weak learner, and $\Omega(f_k)$ is a regularization term that penalizes the complexity of the model.

The XGBoost algorithm optimizes the objective function by iteratively learning the weak learners, using a second-order Taylor expansion to approximate the loss function, and employing regularization terms to avoid overfitting. The optimization can be represented by the following equation:

$$\phi^* = \arg \min_{\phi} \left[\sum_i l(y_i, \hat{y}_i - 1 + \phi f_k(x_i)) + \sum_k k\Omega(f_k + \phi f_k) \right]$$

The execution of the XGBoost classifier requires the use of GridSearchCV.

2.2.7 Shap values

SHAP (SHapley Additive exPlanations) values are a unified measure of feature importance that allows for the explanation of any machine learning model. Developed by Lundberg and Lee (2017), SHAP values provide a way to understand the contribution of each feature to the prediction of a specific instance. These values are based on Shapley values from cooperative game theory, which assign a value to each player in a game that represents their contribution to the overall outcome. In machine learning, each feature is considered a player, and the prediction is the overall outcome. The SHAP values indicate the average marginal contribution of each variable, and they guarantee that the sum of all the feature importance values equals the difference between the model's output and the expected value of the output.

SHAP values have several advantages over other feature importance measures. Firstly, they are model-agnostic, meaning they can be used to explain any machine-learning model. Secondly, they are consistent with the model's underlying assumptions, such as linearity or non-linearity. Thirdly, they explain each instance, not just average feature importance across all instances. Finally, they also provide a measure of uncertainty, indicating the degree of feature importance variation across different instances (Lundberg & Lee, 2017).

The SHAP library in Python is an open-source library for computing SHAP values for various machine-learning models (Strumbelj & Kononenko, 2014). It has been widely used in many fields, such as finance, healthcare, and computer vision. The library can compute SHAP values for a specific instance, a group of instances, or a dataset as a whole. It also provides various visualization tools to help understand the results.

3 Results

3.1 Exploratory analysis

Prior to engaging in data manipulation through dimensionality reduction and subsequent model construction, it is imperative to gain a thorough understanding of the data set to be utilized. To achieve this objective, descriptive statistics were employed. The objective of this methodology is to visually depict the trends of key variables to comprehend the distribution of the data and identify any potential disparities. Additionally, it serves to verify any assumptions made regarding certain variables, as well as to determine the relationship between features and between the target and explanatory variables.

3.1.1 Malnutrition levels

In this section, we will examine the distribution of the variable malnutrition level, without taking into account relationships with other variables. This is a crucial first step to identifying patterns and trends within the data to better contextualize findings.

The below pie chart represents the proportions of each level of malnutrition:

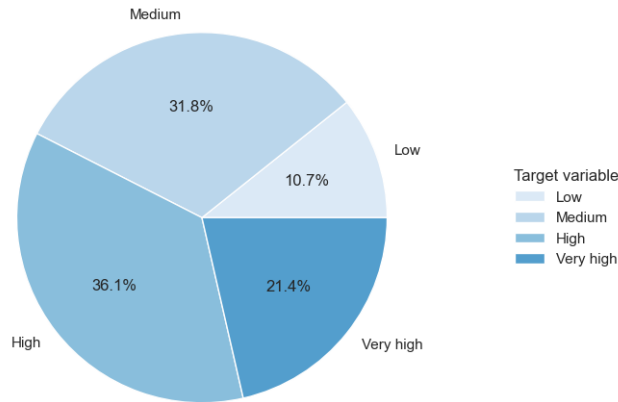
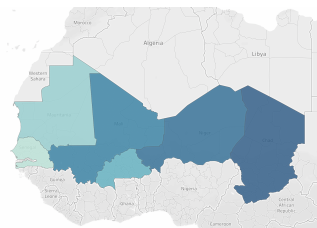


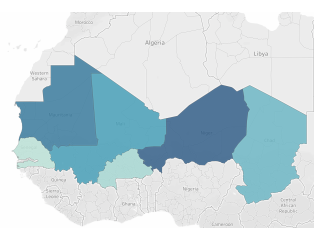
Figure 2: Pie Chart Representation of target variable

These percentages provide an initial understanding of the prevalence of each malnutrition level within the studied population. High level of malnutrition is the most common, affecting slightly more than a third of the population. Medium level of malnutrition is the second most prevalent category, representing nearly a third of the population. Very high level of malnutrition affects over one-fifth of the population, while the lowest level of malnutrition is the least common, accounting for just over 10% of the population.

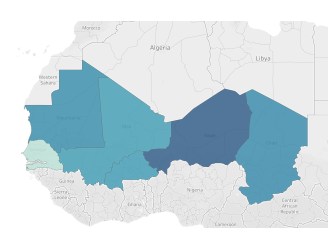
We also wanted to visualize the evolution of the children's malnutrition in the countries, over time, to learn about the trends and to know which countries remain after the database cleaning we did at the beginning.



(a) Average malnutrition level per country (2020)



(b) Average malnutrition level per country (2021)



(c) Average malnutrition level per country (2022)

With the data cleaning, we went from eight countries in the original dataset, to six in the final

one. We can see that countries' situation evolved over the years, for instance that of Mali and Chad improved, while that of Mauritania and Niger got worse.

3.1.2 Explanatory variables

An exhaustive analysis of the summary statistics for the explanatory variables can provide essential insights into the factors contributing to malnutrition. These variables include internally displaced persons (idps), under-5 mortality rate (u5_mortality_rate), access to health care, food insecurity probability, physical infrastructure, and diarrhea prevalence in children under 5 (diarrhea).

Variables	Count	Mean	St. Dev.	Min	25%	50%	75%	Max
idps	1125	4.61	3.0	0.0	4.2	4.54	5.16	12.64
u5_mortality_rate	1125	2.72	3.44	0.0	0.43	1.13	3.57	16.2
Access_to_health_care	1125	6.89	1.31	4.9	5.6	6.8	8.1	9.9
Food_Insecurity_Probability	1125	2.52	1.31	0.0	1.6	2.5	3.6	5.4
Physical_infrastructure	1125	7.37	2.19	0.7	5.6	7.7	9.5	10.0
diarrhea	1125	24.37	21.16	0.0	15.3	18.0	19.8	91.0

Table 1: Summary statistics for some explanatory variables

The average number of internally displaced persons is 4.61, with a standard deviation of 3, indicating a moderately dispersed distribution. The minimum value is 0, and the maximum value is 12.64. This suggests that some areas experience a high concentration of displaced persons, which could strain local resources and contribute to malnutrition.

The mean under-5 mortality rate is 2.72, with a standard deviation of 3.44, revealing a high level of variability across the studied population. The minimum value is 0, while the maximum is 16.2, signifying that some regions may have substantially higher child mortality rates. It could reveal poorer access to healthcare, nutrition, and sanitation.

The average access to health care score is 6.89, with a standard deviation of 1.31. The minimum score is 4.9, and the maximum is 9.9. The relatively low standard deviation indicates a somewhat uniform distribution of healthcare access, though disparities do exist, which may affect the prevalence of malnutrition.

The mean food insecurity probability is 2.52, with a standard deviation of 1.31. The minimum value is 0, and the maximum value is 5.4. These values suggest that food insecurity is a concern in the studied population and could contribute significantly to malnutrition.

The average physical infrastructure index is 7.37, with a standard deviation of 2.19. The minimum score is 0.7, and the maximum score is 10. The wide range of values indicates significant differences in infrastructure quality across the population, which may influence access to essential services, including healthcare, food, and clean water.

The mean diarrhea prevalence is 24.37, with a standard deviation of 21.16. The minimum value is 0, and the maximum value is 91. The high standard deviation and wide range of values suggest that some regions have a much higher prevalence of diarrhea, which can contribute to malnutrition by hindering nutrient absorption and leading to dehydration.

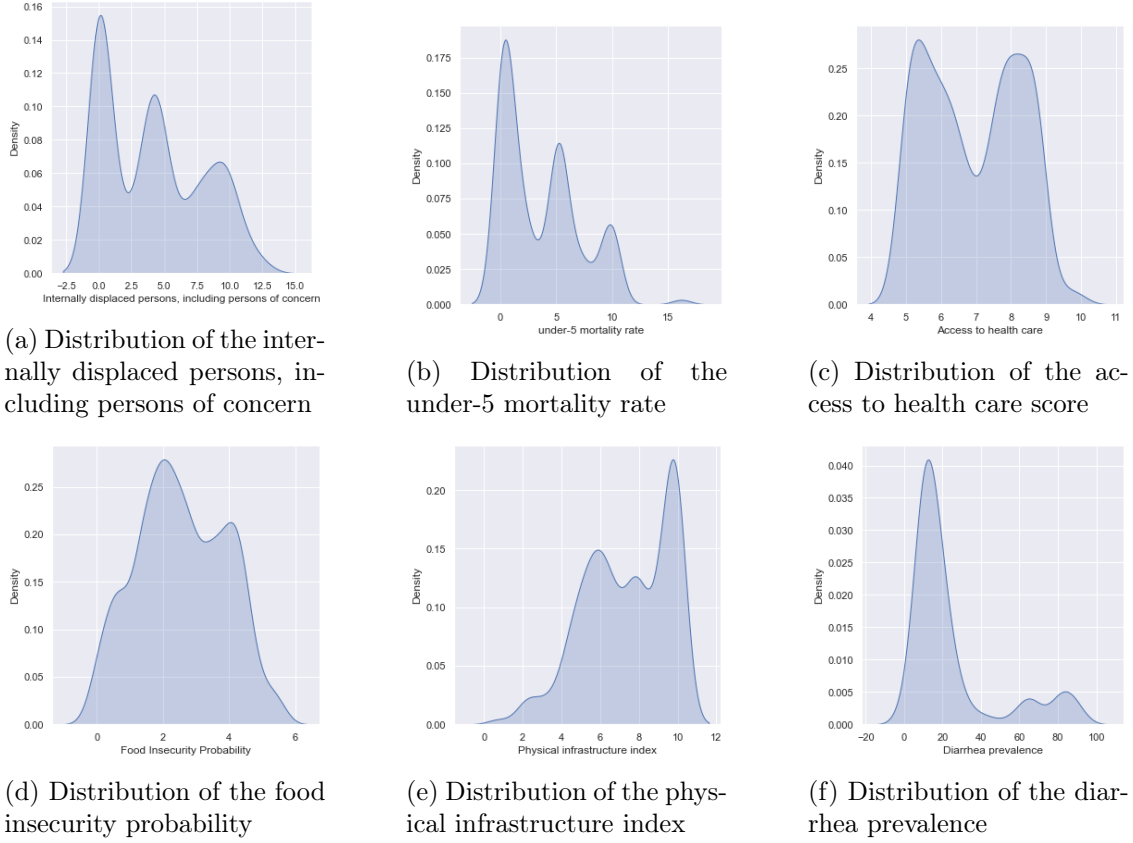


Figure 4: Distributions of some explanatory variables

In conclusion, the summary statistics for the explanatory variables highlight various factors that may influence the different levels of malnutrition observed in the population. To better understand these relationships, we will examine the correlations between these explanatory variables and the target variable, malnutrition levels.

3.1.3 Exploring Correlations Between Explanatory Variables and Malnutrition Levels

We used the ANOVA (Analysis of Variance) method to analyze the relationship between some quantitative explanatory variables and the categorical target variable. The ANOVA test determines whether there is a significant difference in the means of the continuous variable across the categories of the target variable. The results are summarized in the table below:

Explanatory Variable	F-statistic	p-value
idps	56.26	0.00000
u5_mortality_rate	5.11	0.00162
Access_to_health_care	26.49	0.00000
Food_Insecurity_Probability	137.96	0.00000
Physical_infrastructure	24.13	0.00000
diarrhee	0.64	0.58653

Table 2: ANOVA results for the relationship between explanatory variables and malnutrition levels

The ANOVA results indicate that Food Insecurity Probability has the highest F-statistic (137.96), with a p-value < 0.00001 , suggesting a significant relationship with malnutrition levels (this is a natural conclusion). Similarly, IDPs, Access to Health Care, and Physical Infrastructure variables

exhibit important relationships with malnutrition levels, with F-statistics of 56.26, 26.49, and 24.13, respectively, and p-values < 0.00001 .

The under-5 mortality rate variable also significantly correlates with malnutrition levels (F-statistic = 5.11, p-value = 0.00162). However, diarrhea prevalence in children under five has no significant relationship with malnutrition levels (F-statistic = 0.64, p-value = 0.58653).

In summary, the ANOVA analysis suggests that Food Insecurity Probability, IDPs, Access to Health Care, Physical Infrastructure, and under-5 mortality rate are relevant factors related to malnutrition levels in the studied population.

In addition to the previous analyses, we can investigate the distribution of specific variables concerning malnutrition levels. One such variable of interest is 'Physical exposure to flood,' which represents the level of exposure to flood hazards in a given location.

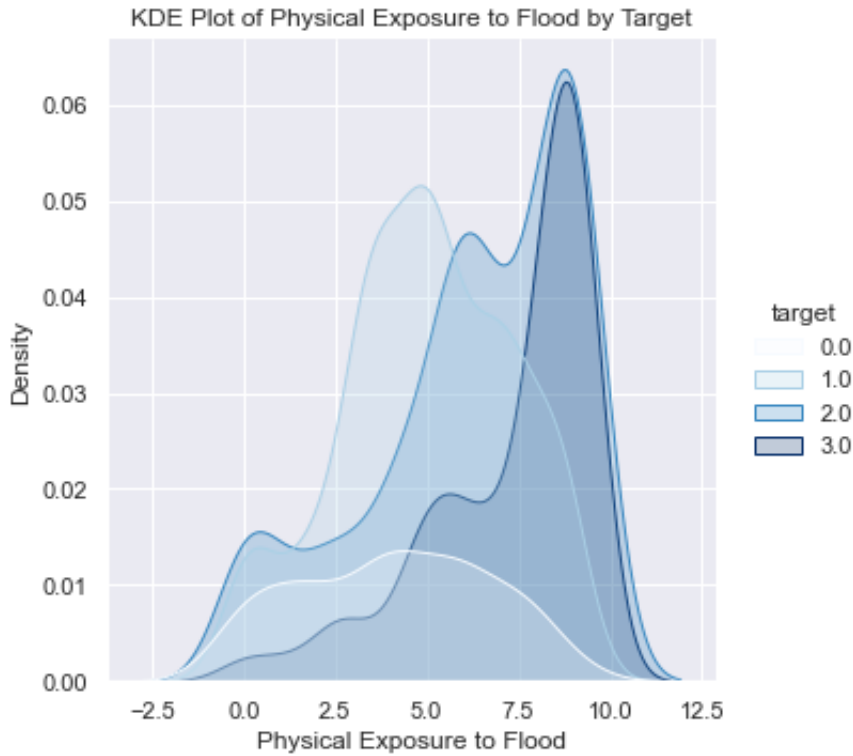


Figure 5: Distribution of the physical exposure to flood in respect of the malnutrition levels

We can make several observations. Locations with higher physical exposure to flood hazards tend to have higher malnutrition levels. Conversely, areas with lower flood exposure generally exhibit lower levels of malnutrition. This trend suggests a potential link between the vulnerability to flood hazards and the prevalence of malnutrition in the studied population.

3.2 Predictive analysis results

3.2.1 Dimensionality reduction

In this research stage, we intended to reduce the dimensionality of the cleaned dataset to improve the performance and interpretability of the models that would be used in the next phases. We began by computing the correlation between the variables in the dataset. We discovered that 22.2 percent of the variables had a correlation greater than 80%. A strong correlation between variables can result in multicollinearity, which can harm model performance and interpretability. We proposed using two dimensionality reduction approaches to handle this problem: Lasso and Elastic-net.

According to Hastie (2005), Lasso is a linear model that adds a regularization component to the cost function, a sum of the coefficients' absolute values. This regularization term is governed by α , a hyperparameter that sets the intensity of the regularization. Lasso has the benefit of being able to reduce some coefficients to zero, thereby eliminating them from the model. This model is useful when there are numerous variables, but just a subset is important to the model. However, if the regularization is too strong, it might lead to underfitting since it removes important variables from the model. Elastic-net is a combination of Lasso and Ridge regularization. Like Lasso, it shrinks some coefficients to zero but also shrinks all coefficients by the same amount, preventing overfitting. Elastic-net is controlled by two hyperparameters: α and l1_ratio . α controls the strength of the regularization, and l1_ratio controls the proportion of L1 regularization in the combination. This combination of L1 and L2 regularization can be useful when the data contains a mix of correlated and uncorrelated variables (Hastie, 2005).

Before running Lasso and Elastic-net, we scaled the data. Scaling the data is essential because these methods are sensitive to the scale of the variables. By scaling the data, we ensure that all variables are on the same scale, allowing regularization to be applied uniformly across all variables and preventing some variables from dominating the model.

We used the GridSearchCV function from the scikit-learn library to optimize the hyperparameters of the Lasso and Elastic-net models. GridSearchCV is a powerful method that performs an exhaustive search over a specified parameter grid, and it selects the best set of hyperparameters based on cross-validation performance. We optimized the hyperparameters ' α ' for Lasso and ' α ' and ' l1_ratio ' for Elastic-net. We fitted the GridSearchCV objects to the data, and the best hyperparameters selected by the grid search are shown in Table 1.

Table 3: Best hyperparameters selected by the grid search

Hyperparameter	Lasso	Elastic-Net
α	0.01	0.1
L1_ratio		0.09

We checked the percentage of coefficients shrunk to zero; 42.7% in the case of Lasso and 26.9% for Elastic-net. We then compared the Lasso and Elastic-Net models using cross-validation and R-squared. The results showed that Elastic-net was slightly better, with a mean cross-validation score of 0.52 and an R-squared of 0.75. We proceeded in the following sections with Elastic-net due to its considerable advantages over Lasso and its ability to handle correlated variables effectively.

As a result of the Elastic-net approach, 41 variables were shrunk to zero, leaving 111 variables in the dataset.

3.2.2 Machine learning algorithms

To ensure a fair evaluation of the models, the dataset was split into two sets: a training set and a test set. The training set contained 80% of the data and was used to train the models with the selected hyperparameters. The remaining 20% of the data was used as the test set, which allowed us to assess the performance of the models. By comparing the predictions of our models against the actual values in the test set, we could evaluate how well our models performed. Since the training set is larger than the test set, it provides enough data for the models to be trained accurately, while the test set ensures that the models generalize well to new, unseen data.

(a) Random Forest Classifier

The Random Forest Model is estimated with hyperparameters tuning and the 'RandomForest-Classifer' package of scikit-learn in Python. The hyperparameters that are sought to be adjusted in this model are the following:

- $n_estimators$: refers to the number of trees to be estimated in the Random Forest.

- `max_features`: refers to the maximum number of features considered at every split.
- `max_depth`: refers to the maximum number of the levels in each tree.
- `criterion`: refers to the function to measure the quality of a split.

We employed a grid-search strategy utilizing 3-fold cross-validation to select appropriate hyperparameters. While increasing the number of iterations and cross-validation folds may yield more comprehensive results and reduce the risk of overfitting, it is important to consider the trade-off with computational time. The grid search results are presented in the table of selected hyperparameters.

Table 4: Best hyperparameters selected by the grid search

Hyperparameter	Grid values	Selected value
<code>n_estimators</code>	200, 500	200
<code>max_features</code>	'auto', 'sqrt', 'log2'	'auto'
<code>max_depth</code>	4,5,6,7,8	8
<code>criterion</code>	'gini', 'entropy'	'gini'

(b) Support Vector Classifier

The second method we decided to implement for this prediction study is the Support Vector Machine (SVM). Although time-consuming when it is run, this model can be exceptionally performing and offers multiple options through hyperparameter tuning. As for the Random Forest model, the initial dataset has been split as follows: 80% for the training set and 20% for the test set.

The Support Vector Classifier (SVC) of scikit-learn proposes multiple parameters, including:

- **Kernel**: gives the kernel that will be used in the model. The kernel is the mathematical function that will take the data as input and provide an output that will be used to make predictions. The kernel can be linear or non-linear. We chose the second option because it is more adaptive to the data. We would have tried with a radial basis function (gaussian) kernel and a polynomial, but we restricted ourselves to an RBF for computing capacity reasons.
- **C**: cost parameter, makes a tradeoff between model accuracy and ability to predict unknown data. The higher C, the more permissive the model.
- **Gamma**: non-linear regressor parameter. The higher Gamma, the smaller the variance of the model.

The `gridsearchCV` was applied to select the best hyperparameters, and the selected values are presented in the table below.

Table 5: Best hyperparameters selected by the grid search

Hyperparameter	Grid values	Selected value
<code>C</code>	0.01, 0.1, 1, 10, 100, 1000	1000
<code>Gamma</code>	1, 0.1, 0.01, 0.001, 0.0001	0.1

(c) K-Nearest Neighbors

For the KNN algorithm, we set up a range of values for the parameter `n_neighbors`, which specifies the number of neighbours to consider when making a prediction. The `GridSearchCV` function is then used to perform a grid search over a range between 1 and 31 `n_neighbors`, using 10-fold cross-validation to evaluate the performance of each combination of hyperparameters. The value selected

from the grid was 1, which was then applied to the KNeighborsClassifier function from the scikit-learn package.

(d) eXtreme Gradient Boosting

The last model we applied to in this report is eXtreme Gradient Boosting. The package for implementing this algorithm includes several parameters, among them we can mention:

- `learning_rate`: it shrinks the contribution of each tree.
- `max_depth`: refers to the maximum number of the levels in each tree.
- `min_child_weight`: minimum sum of instance weight (hessian) needed in a child.
- `colsample_bytree`: is the subsample ratio of columns when constructing each tree.

We applied the grid values in the table below in the GridSearchCV with 3-fold cross-validation, and the best hyperparameters were selected. These were later used in the XGBoostClassifier algorithm.

Table 6: Best hyperparameters selected by the grid search

Hyperparameter	Grid values	Selected value
<code>learning_rate</code>	0.0001, 0.001, 0.01, 0.1	0.1
<code>max_depth</code>	3, 5, 6, 7	6
<code>min_child_weight</code>	3, 5, 6, 7	3
<code>colsample_bytree</code>	0.1, 0.5, 1.0	0.8

3.2.3 Model selection

The optimal model selection is based on a comprehensive evaluation of multiple performance metrics, including the Mean Squared Error (MSE) and accuracy. The values obtained for the four models are noted in the table below:

Table 7: Model selection

Model	MSE (test sample)	Accuracy
Random Forest	0.147	0.866
Support Vector Classifier	0.189	0.836
K-Nearest Neighbors	0.189	0.861
eXtreme Gradient Boosting	0.118	0.895

According to these two metrics, the most efficient model is the XGBoost since it obtains the lowest MSE (0.118) and the highest accuracy score (89.5%). The second leading is the Random Forest Classifier, the third best is the K-NN and the last is the SVM model.

An effective method to evaluate the performance of our classification models is the AUC ROC (Area Under the Curve Receiver Operating Characteristics) plot. It helps to visualize if the model is able to distinguish between classes, by measuring sensitivity (or true positive rate) and specificity (or false positive rate). In our case of multi-class classification, the "one-versus-rest" methodology is applied for each of the four classes.

We plotted the AUC ROC plot for the two models that showed the best accuracy scores: the Random Forest Classifier and the Extreme Gradient Boosting:

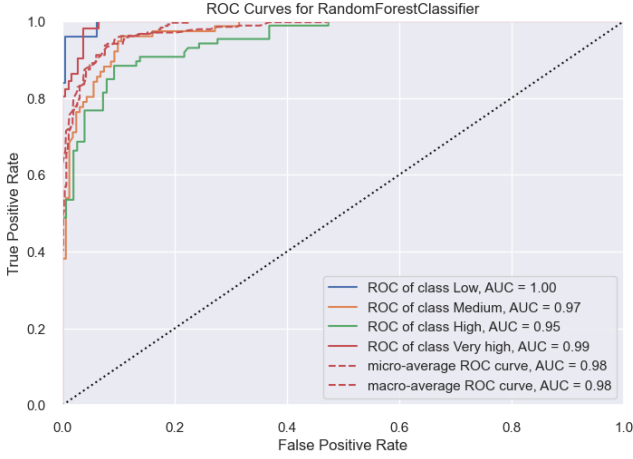


Figure 6: AUC ROC Plot for Random Forest Classifier

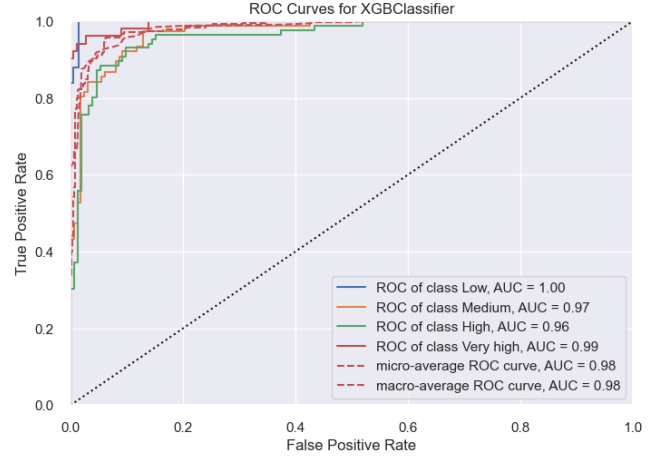


Figure 7: AUC ROC Plot for XGBoost Classifier

Both models are extremely performing for the 'Low' class since they can distinguish this class from the others 100% of the time. Both models have equivalent performances on the 'Medium' and 'Very high' classes, whereas, on the 'High' class, the XGBoost model stands out.

In order to study in more detail the prediction performance of the XGBoost model, we constructed its confusion matrix:

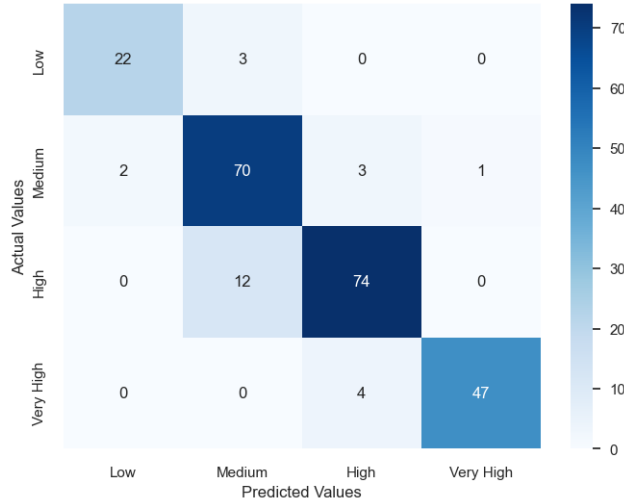


Figure 8: Extreme Gradient Boosting confusion matrix

The XGBoost model has satisfying results on every class of the test sample, except for the 'Medium' class, where 15% has been predicted as 'High'.

3.2.4 SHAP Values for XGBoost

SHAP values, or SHapley Additive exPlanations, explain the output of any machine learning model. They provide interpretable and locally accurate importance scores for each feature by computing the contribution of each variable to the prediction of a given instance. They are based on Shapley values from cooperative game theory and can be used to explain global and local forecasts. (Lundberg and Lee, 2017)

In this study, we use SHAP values to provide interpretability and assess the impact of each variable on the prediction. This approach allows us to offer a comprehensive and useful analysis for policymakers, as they will better understand the most important factors in determining the level of malnutrition.

We computed SHAP values using the SHAP library in Python. We first created an instance of the Explainer class, passing in the XGBoost model and the training data. Finally, we used the explainer to compute the SHAP values for the test data. In the following, we will present global and individual plots and the most valuable insights we can obtain for them.

(a) Global plots

1. Variable Importance Plot.

The Variable Importance Plot provides a visual representation of the relative importance of each feature in the classes. The plot displays the features in descending order of importance, measured by their average absolute SHAP values.

These results can be used to identify potential areas for intervention and improvement. The plot suggests that access to health coverage, food insecurity, the death rate associated with the number of deaths caused by malaria, and whether the person is internally displaced or if it belongs to a vulnerable group are crucial in addressing malnutrition, especially in the case where the level of malnutrition is very high.

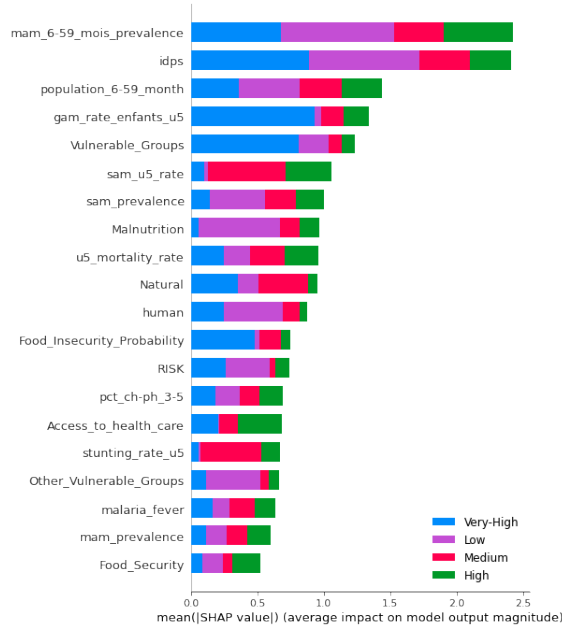


Figure 9: Variable Importance Plot

2. Beeswarm Summary Plot.

The Beeswarm Summary plot presents an information-dense summary of how the top features in a dataset impact the model's output for a specific class, specifically for the high level of malnutrition class in the case of the figure below. The y-axis displays the names of the variables, while the x-axis indicates the SHAP values. The plot uses a colour coding system: red represents high feature values, and blue represents low feature values. It reveals, for example, that a high food insecurity probability increases the predicted level of malnutrition.

The results of the Beeswarm summary plot align well with the reality and the characteristics of how one would expect the features to impact the level of malnutrition. Factors such as being in a vulnerable group or being internally displaced persons (IDPs), having a higher food insecurity and droughts probability, and a higher mortality rate are commonly associated with a higher level of malnutrition. These individuals are more likely to lack access to basic necessities such as health coverage and food security and may be more prone to poverty and malnutrition. On the

other hand, factors such as access to health care are commonly associated with a lower level of malnutrition.

The Beeswarm summary plot can identify potential areas for intervention and improvement and inform policies and programs to address this issue, especially for the "high-level of malnutrition" class.

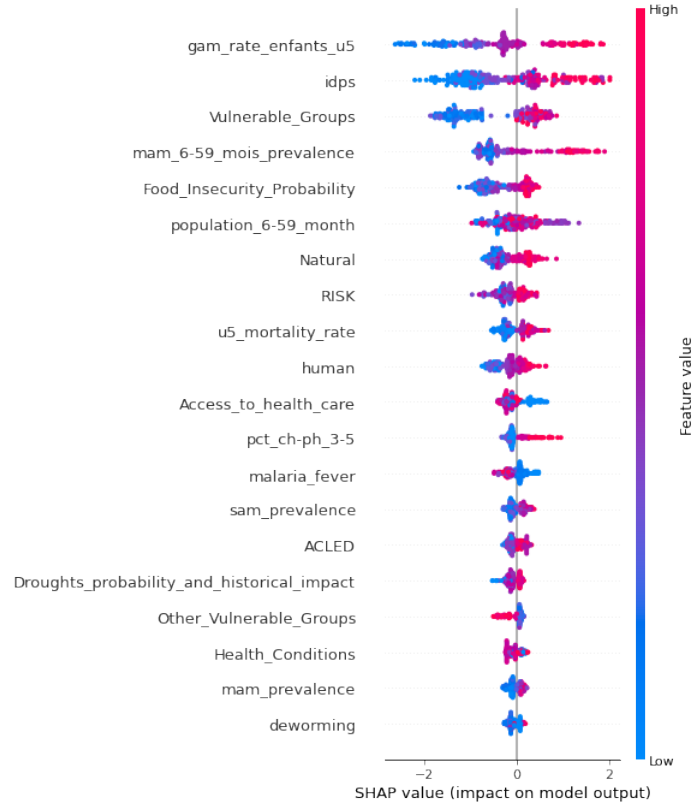


Figure 10: Beeswarm Summary Plot for class "High level of malnutrition"

(b) Individual plots

The Force plot and Waterfall plot are visual representations of which features most influenced the model's prediction for a single individual. They show the importance of considering individual-specific factors when addressing malnutrition, as the impact of certain variables may differ from one individual to another. In both plots, the essential features to making the prediction are shown in red and blue, with red representing features that pushed the model score higher and blue illustrating features that moved the score lower.

The Force plot shows the features that had a higher impact on the score closer to the dividing boundary between red and blue, and the bar represents the size of that impact. On the other hand, the Waterfall plot shows the feature's importance as a stacked bar chart. It provides an easy way to identify each feature's contribution score by showing its cumulative effect on the final prediction. Together, these two plots offer a comprehensive view of the feature importance for a specific case and allow the identification of critical factors driving the prediction.

The baseline prediction for this individual is -0.614, but after considering the feature values, the final prediction is -2.253. For this individual, these plots highlight the key factors that determine its final prediction, which can be used to identify potential areas for intervention and improvement for this individual.

The plot indicates the most critical features: food security, IDPs, and deworming. They negatively contribute to the final prediction, with values of -0.65, -0.49, and -0.33, respectively. These

findings suggest that even though these factors may not be enough to lift an individual out of malnutrition, they could help improve their situation.

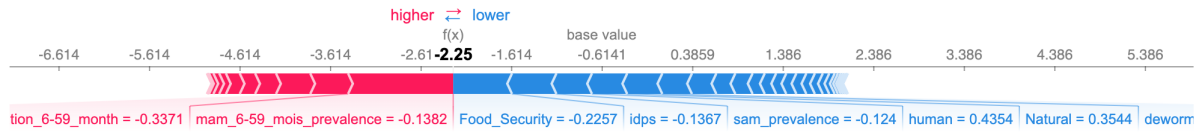


Figure 11: Force Plot for one individual

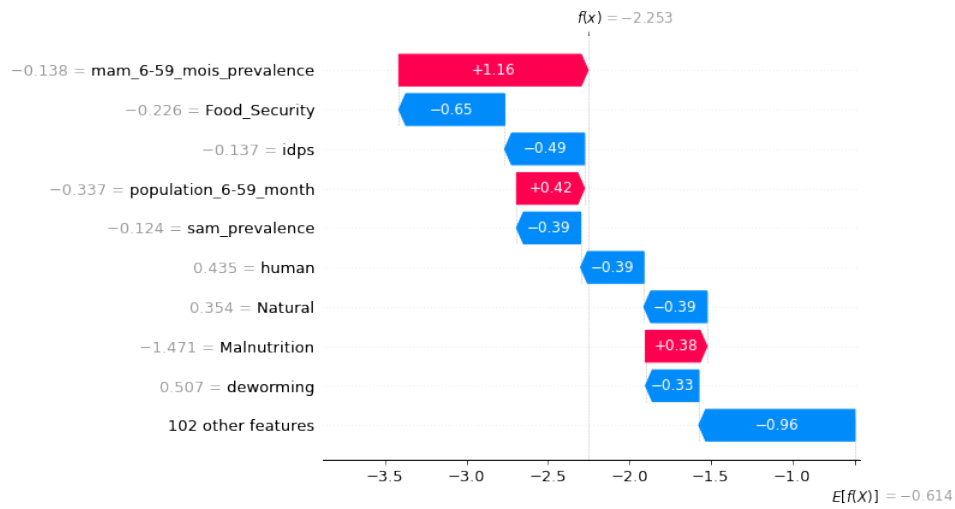


Figure 12: Waterfall Plot for one individual

4 Conclusion

In this study, we aimed to forecast malnutrition in children aged between 6-59 months from Senegal and the G5 Sahel countries: Burkina Faso, Chad, Mali, Mauritania, and Niger. The strategy was first dealing with the missing values through a Hot-Deck with KNNs. Following, we performed two dimensionality reduction methods, Lasso and Elastic-net, where the second proved to be better. We then proposed four machine learning models widely used in the most recent literature: Random Forest, KNN, Support Vector Machine, and XGBoost. Our findings showed that the eXtreme Gradient Boosting model performed best in prediction accuracy, with the lowest mean squared error (MSE) compared to the rest of the models. Additionally, the ROC curves proved how the XGBoost model stands out compared to the second-best predictor, Random Forest. Finally, the confusion matrix shows satisfying results on every class of the test sample. Furthermore, the analysis of SHAP values revealed that variables related to access to health care, food insecurity, internally displaced persons, and the death rate associated with malaria were among the most important predictors of the level of malnutrition. We also illustrated the features that most influenced the model prediction for one individual, allowing us to identify the different drivers in particular cases.

As for possible future work, one possibility would be to incorporate additional variables or try another dimensionality reduction approach into the analysis to improve the accuracy of predictions. Another option would be to test different machine learning models further to see if any other model outperforms XGBoost, like Neural Networks, which is also widely used in the literature. Also, we could shrink the database to keep only the most relevant variables selected in this report and use them to run the four machine learning models again to see if the accuracy improves. Additionally, we could extend the study to other countries with similar characteristics to test if our findings can be generalized.

5 References

Ashis Talukder, Benojir Ahammed, (2020) Machine learning algorithms for predicting malnutrition among under-five children in Bangladesh, *Nutrition*, Volume 78, 110861, ISSN 0899-9007, <https://doi.org/10.1016/j.nut.2020.110861>

Bitew, F., Sparks, C., & Nyarko, S. (2022). Machine learning algorithms for predicting undernutrition among under-five children in Ethiopia. *Public Health Nutrition*, 25(2), 269-280. doi:10.1017/S1368980021004262

Fenta, H.M., Zewotir, T. & Muluneh, E.K. (2021) A machine learning classifier approach for identifying the determinants of under-five child undernutrition in Ethiopian administrative zones. *BMC Med Inform Decis Mak* 21, 291. <https://doi.org/10.1186/s12911-021-01652-1>

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning with Applications in R*. Springer Texts in Statistics. ISBN 978-1-4614-7137-0, doi: 10.1007/978-1-4614-7138-7

Joenssen, D.W., Bankhofer, U. (2012). Hot Deck Methods for Imputing Missing Data. In: Perner, P. (eds) *Machine Learning and Data Mining in Pattern Recognition*. MLDM 2012.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17)*. Curran Associates Inc., Red Hook, NY, USA, 4768–4777.

Kwon, Soonchang, (2014) Imputation of Missing Data Based on Hot Deck Method Using K-nn, *Journal of Information Technology Services*, vol. 13, no. 4, pp. 359–375.

Md Nafiul Alam Khan, Rossita Mohamad Yunus, (2023) A hybrid ensemble approach to accelerate the classification accuracy for predicting malnutrition among under-five children in sub-Saharan African countries, *Nutrition*, Volume 108, 111947, ISSN 0899-9007, <https://doi.org/10.1016/j.nut.2022.111947>.

Md. Merajul Islam, Md. Jahanur Rahman, Md Moidul Islam, Dulal Chandra Roy, N.A.M. Faisal Ahmed, Sadiq Hussain, Md Amanullah, Md. Menhazul Abedin, Md. Maniruzzaman, (2022) Application of machine learning based algorithm for prediction of malnutrition among women in Bangladesh, *International Journal of Cognitive Computing in Engineering*, Volume 3, Pages 46-57, ISSN 2666-3074, <https://doi.org/10.1016/j.ijcce.2022.02.002>.

Rahamuddin Khan, Manish Ahirwar, Piyush kumar shukla (2019) Predicting Malnutrition Disease Using Various Machine Learning Algorithms in *International journal of scientific & technology research*, ISSN 2277-8616

Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society. Series B: Statistical Methodology*, 67(2), 301-320. <https://doi.org/10.1111/j.1467-9868.2005.00503.x>